

Exploring India's Coral Reefs: A Comprehensive Analysis of Biodiversity, Threats, Conservation, Restoration, and Monitoring

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Abstract

Coral reefs represent some vital unique marine ecosystems globally. Apart from providing several ecologic services and food and economic benefits, coral reefs are stunning underwater scenes that offer a variety of leisure opportunities and raise the nation's appeal to both domestic and foreign tourists. Global environmental degradation and climate change

are placing enormous strain on coral reef ecosystems. The exclusive economic zone (EEZ) of Indian coastal waters covers 2.02 million km² and a coastline spanning over 8000 km. However, the reef formation is found only across the Gulf of Mannar (GoM), the Andaman and Nicobar Islands (ANI), the Gulf of Kachchh (GoK), and the Lakshadweep Islands (LaI). Patchy reefs are reported from Ratnagiri and Malvan coasts. Coral reefs of each of these areas have a unique form or structure. The estimated area under coral reef in India is 2375 km². The Indian Coastal Regulation Zone (CRZ) Notification of 1991 safeguards all coral reefs in India. The CRZ1 category encompasses all coral reef habitats. Additionally, national, state, and non-governmental organizations in India have started taking steps towards artificial reef deployment, reef transplantation and restoration. This chapter discusses the state of coral reefs in India and the emerging concerns primarily based on satellite remotely sensed observations, supplemented by data from other sources. Besides, there is an attempt to deliberate on some of the management issues to preserve and restore coral reefs in India. This includes the successful application of the novel eDNA metabarcoding in coral reef

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restoration activities, which would support the creation of programs for fair conservation and data-driven coastal zone management.

Keywords

India's coral reefs · Biodiversity · Satellite observation · Coastal regulation zone · Metabarcoding

27.1 Introduction

The ocean supports life on Earth and plays a crucial role in regulating the global climate. Among the most remarkable marine ecosystems worldwide are coral reefs, characterized by colonies of coral polyps intertwined through calcium carbonate structures in surface or shallow waters. The ability of reefs to build corals engages in a synergetic relationship with photosynthetic algae known as zooxanthellae, residing in their tissues and facilitating mutualistic interactions. Coral reefs' substantial role is absorbing significant amounts of wave energy, providing essential protection against flood events and hurricanes for coastal communities, lives, and properties. Moreover, these ecosystems contribute significantly to fisheries, acting as crucial nurseries and habitats that support the livelihoods of fishermen and related industries by supplying food and income. Beyond their economic importance, coral reefs enhance the aesthetic appeal of marine environments, offering captivating seascapes that facilitate various recreational events. Additionally, the presence of vibrant coral reefs enhances the allure of a country for international tourism markets, showcasing the ecological and aesthetic value of these marine wonders.

According to the International Union for Conservation of Nature (IUCN), globally, among all ecosystems, coral reefs host the greatest biodiversity and also offer direct support to over 500 million people worldwide, predominantly in economically challenged nations. These reefs are astounding, visually stunning, biologically diverse, invaluable and most productive ecosys-

tems on Earth. The UN Environment Report 2018 estimated that the global coral reef value is 375 billion US\$ per year considering goods and services.

India boasts a coastline spanning approximately 8000 km and features diverse types of reefs, with an average area of 2375 km², with a primary concentration in four key regions: the Gulf of Mannar (GoM), Andaman and Nicobar Islands (ANI) in the Bay of Bengal, Gulf of Kachchh (GoK) and Lakshadweep Islands (LaI) in the Arabian Sea (Baswapoor and Irfan 2018). Based on remote sensing data, the total area under coral reefs has been estimated as 841 km², 2384 km², and 1420 km² during the periods 1992–1993 (Garg et al. 1998), 2004–2007, and 2007–2008, respectively (SAC 2011; Bahuguna et al. 2013). Also, the estimation by Spalding et al. (2001) revealed a total coral reef coverage of 5790 km², which accounts for 2.04% of global coral coverage. Subsequent estimations of coral reef coverage in 2009 (SoE 2009) and early 2010 (Venkataraman 2011; Bahuguna et al. 2013) indicated a significant decline in coral reefs, with coverage reduced to approximately 2375–2383 km², equal to 0.92% of global coral coverage (Souter et al. 2021). Bahuguna et al. (2013) estimate the coral coverage in India, with significant area observed in the Andaman and Nicobar Islands (~1021 km²) and the Lakshadweep Islands (~934 km²). The Gulf of Kachchh has approximately 352 km² of coral coverage, while the Gulf of Mannar/Palk Bay has about 76 km². The Andaman and Nicobar Islands and the Lakshadweep, together, account for around 40% of India's total reef area (Saroj et al. 2016).

Four major types of coral reefs found in Indian sub-continent are:

27.1.1 Platform Reefs

These reefs have no lagoon and are almost flat. They are found between a shore and a barrier reef or near atolls, usually on the shallower parts of the continental shelf. Platform reefs are very common in the Indian Gulf of Kachchh.

27.1.2 Fringing Reefs

In India, the fringing reefs are dominating in the ANI, covering an expanse of 1021.46 km². Such reefs adhere directly to a shore or border in shallow waterways or lagoons. Moreover, some fringing reefs are found in the GoM and the GoK, with respective areas of 352.50 km² and 75.93 km² (De et al. 2017).

27.1.3 Barrier Reefs

Barrier reefs are observed mostly in the ANI of India. They are formed when a deep canal or lagoon divides an island's shoreline from that of a mainland. Notably, the best example of a barrier reef outside India is the Great Barrier Reef in Australia, which is distinguished by its separation from the mainland by vast stretches of water while paralleling the coastline. In India, barrier reefs can be found in the GoM and the ANI (Saroj et al. 2016; De et al. 2017).

27.1.4 Atolls

An atoll is a circular coral formation that often supports low-lying islands and surrounds a lagoon. In India, atolls are chiefly found in the

Lakshadweep Islands, with an area of 933.7 km², characterized by an unevenly circular ring of reefs encircling a lagoon and a low-lying island is typical. This reef type is common in the Indian and South Pacific Oceans (Saroj et al. 2016).

27.1.5 Patchy Reefs

Patchy reefs are often encountered on the continental shelf between Vengurla and Vijaydurg along the Maharashtra coast, as well as in intertidal zones like Ratnagiri, Malvan, and Redi. Additionally, submerged patchy reefs may be found adjacent to the western coast, like the Angria Bank off Sindhudurg in Maharashtra, Grande Island in Goa, Netrani Island on the Karnataka coast, Gaveshana Bank off the Malpe coast, Kachchh's of Gujarat (most northerly coral reefs), and near Quilon and Vizhinjam along the Kerala coast. Additionally, Enayem along the Tamil Nadu coast hosts these reefs. Mesophotic coral ecosystems are situated off the Puducherry coast (Jasmine et al. 2009; De et al. 2017; Kumar et al. 2019; Laxmilatha et al. 2019).

These coral reefs not only provide support to biodiversity but also hold immense value to humankind. They play a crucial role both ecologically and economically (Table 27.1).

Table 27.1 Economic importance of coral reefs besides their ecosystem function

Type of services	Function	Description
Provisional services	Medicinal properties	Coral reefs have the potential to cure diseases such as ulcers, leukaemia, and lymphoma (Baswapoor and Irfan 2018). Some genera of corals such as <i>Sinularia</i> , <i>Echinomuricea</i> , <i>Paraminabea</i> , etc. have anti-inflammatory properties (Cooper et al. 2014)
	Food	Coral reefs are crucial fish nurseries, supporting 25% of ocean fish and providing essential protein, which attracts national and international fishing companies, significantly contributing to revenue (Baswapoor and Irfan 2018)
	Rich in minerals	Abundant in limestone and calcium-rich coral sand (Baswapoor and Irfan 2018).
Regulating services	Protects coastline	The worldwide coral ecosystem protects the shoreline by absorbing 97% of the energy from incoming waves. This protection benefits around 197 million individuals residing within 50 km of reefs and below 10 m in elevation (Ferrario et al. 2014)
	Water quality	Corals and marine sponges are filter feeders, consuming suspended particles in the water. Thus improving the quality and clarity of our coastal waters (UNEP 2020)
Cultural services	Tourism	Research indicates that, on average nations with coral reef industries generate over 50% of their Gross National Product (GNP) from these ecosystems. Additionally, the reefs themselves attract tourists, fostering water sports economies centred around snorkelling and Scuba diving fees (Baswapoor and Irfan 2018)

27.2 Coral Diversity

India is home to approximately 585 coral species, categorized into 23 families and 108 genera, within an area of about 2383 km² (De et al. 2020; Venkataraman 2007). The highest species richness is documented in the ANI, with 523 species belonging to 95 genera and 23 families, followed by 169 species belonging to 46 genera and 16 families from the GoM; 165 species belonging to 54 genera and 17 families from LaI; 76 species of 30 genera and 12 families from Gulf of Kachchh (De et al. 2020). In addition to these reefs, smaller reefs on India's west coast, such as the Malvan Marine Sanctuary, Angria Bank, Grande Island, and Netrani Island, are also known to include

Scleractinian fauna (De et al. 2020) (Figs. 27.1 and 27.2, Table 27.2).

The largest island in the Nicobar group and the southernmost island in the Andaman and Nicobar group, the Great Nicobar Island is part of the UNESCO-MAB-Network of biosphere reserve. Here, the Simpson's density index for coral reefs goes from 0.93 to 0.99, the Shannon-Weiner diversity index ranges from 2.84 to 4.45 and Laxman beach was found to be with most diversified coral reefs. The presence of 29.98% of *Scleractinian* coral species found in Great Nicobar Island denotes the sense of ecological attributes behind the sustainable support towards the coral settlement, growth, and development (Mondal et al. 2016).



Fig. 27.1 Coral reef site at Andaman and Nicobar Island. (Photo Credit @ Author Yogesh Kumar, J.S. ZSI “Coral reef site at ANI”. 2015)



Fig. 27.2 Coral reef site at Lakshadweep Island. (Photo Credit @ Author Yogesh Kumar, J.S. ZSI. “Coral reef site at Lakshadweep Islands”)

Table 27.2 Location and list of coral family and species observed in Indian sub-continent

Key areas	No of species	Coral family	List of common coral species	List of coral species (Ref)
Andaman and Nicobar Islands (A&N Islands)	523	Acroporidae Pocilloporidae Poritidae Faviidae Milleporidae	<i>Acropora</i> , <i>Montipora</i> , <i>Pocillopora</i> , <i>Porites</i> , <i>Goniopora</i> , <i>Favia</i> <i>Millepora</i> , <i>Sarcophyton</i> sp., <i>Lobophytum</i> sp., and <i>Sinularia</i> sp.	Venkataraman (2006)
Lakshadweep Islands	165	Acroporidae Pocilloporidae Euphyllidae Oculinidae Siderastridae Agariciidae Fungidae	<i>Acropora humilis</i> , <i>A. muricata</i> (formerly <i>A. formosa</i>), <i>A. intermedia</i> , <i>A. hyacinthus</i> , <i>Pocillopora verrucosa</i> , <i>Euphyllia glabrescens</i> , <i>Galaxea fascicularis</i> , <i>Psammocora contigua</i> , <i>P. haimeana</i> , <i>Pavona maldivensis</i> , <i>P. clavus</i> , <i>Fungia danai</i> , and <i>Podobacia crustacea</i>	Venkataraman (2006)
Gulf of Mannar (GoM)	169	Acroporidae Pacilloporidae Poritidae	<i>Montipora monasteriata</i> , <i>M. informis</i> , <i>M. spumosa</i> , <i>M. turgescens</i> , <i>M. venosa</i> , <i>M. verrucosa</i> , <i>M. digitata</i> , <i>M. millepora</i> , <i>M. manauliensis</i> , <i>Acropora digitifera</i> , <i>A. secale</i> , <i>A. intermedia</i> , <i>Pocillopora verrucosa</i> , <i>Porites mannarensis</i> , <i>P. exserta</i> , and <i>Goniopora stutchburyi</i> . Species like <i>Montipora millepora</i> , <i>M. jonesi</i> , <i>M. manauliensis</i> , <i>M. edwardsi</i> , <i>M. exserta</i> , <i>Acropora rudis</i> , <i>A. valenciennesi</i> , <i>A. microphthalma</i> , <i>Porites exserta</i> , and <i>Porites mannarensis</i>	Venkataraman (2006)
Gulf of Kachchh (GoK)	76	Anacropora Merulinidae Poritidae Siderastridae	<i>Montipora venosa</i> , <i>Cosinaria monile</i> , <i>Hydnophora excessa</i> , <i>Turninaria petata</i> , <i>Goniastrea pectinata</i> , <i>Platygyra sinensis</i> , <i>Cyphastrea serialia</i> , <i>Porites compressa</i> , <i>Goniopora stutchburyi</i> , <i>Siderastrea savignayana</i> , and <i>Acanthastrea hillae</i>	Venkataraman (2006)

27.3 Threats to Coral Reefs of India

Among the ecosystems that face the greatest threat on Earth are coral reefs. Since the 1980s, there has been a decline in the health of coral reefs in all of the major tropical oceans, with a global average decline in coral cover of 30–50%. Local and regional stresses like pollution, overfishing, and habitat destruction are the main reasons for these decreases. It is generally accepted that the reef's geographical expanse has decreased in recent decades as a result of rising human pressures, even if the entire reef area cannot be accurately assessed (Ramadas and Rajeswari 2011; Rebekah and Inamdar 2018). Especially, severe bleaching can lead to coral mortality, and accordingly it is important to determine how the Sea Surface Temperature (SST) affects the coral reefs (Arora et al. 2019). Global climate change (prolonged and intense El Nino) is recognized as a major factor for the coral reef decline, associated with ocean acidification, outbreak of disease, and bleaching events. Whereas uncontrolled fishing practices and mining of lime and coral were previously considered the primary threats to corals (Edward 2002), additional threats such as nitrogen pollution and seawater warming have emerged as significant factors affecting coral reefs (Painter et al. 2023). The frequency of coral damage brought on by pollution and sewage discharge, damage from bottom trawling, siltation from coastal development, the invasion of exotic species, and increased coral mortality due to increasing algal blooms (Chandrasekaran et al. 2008; Machendiranathan et al. 2016; Asir et al. 2020; Raj et al. 2020). The reef's ecological dynamics have been impacted by the growing amount of physical damage caused by tourists and recreational divers (Purvaja et al. 2019). The coral reefs of Andaman and Nicobar face a range of threats, including natural disasters such as the 2004 tsunami and the 2011 tropical storm. Human activities, particularly tourism and poaching, also pose significant risks (Jeyabaskaran et al. 2007). Climate change-induced events (El Nino), such as the 2010 mass coral bleaching, further exacerbate these threats (Majumdar et al.

2018). Notwithstanding these challenges, the diverse and dynamic coral reefs in the region possess the capacity to act as a refuge in the midst of global warming. Further, rising levels of turbidity, sedimentation, and discharges from cities and industries have been the results of anthropogenic shoreline alteration, heavy industry, and rapid coastal expansion in recent years (MoEF 2014; Magesh and Krishnakumar 2019; Jinoj et al. 2021).

27.4 Impact of Climate Change on Corals of India

Coral reefs are the utmost imperil ecological system on Earth, facing significant threats from unprecedented global warming and climate changes, compounded by escalated local pressures. Coral reef ecosystems are highly susceptible to changes in the physical environment brought about by climate change. Since the 1980s, the process known as “bleaching” of coral reefs, which is caused by unusually high water temperatures, has had a hugely negative impact on the entire world (Baker et al. 2008). UNESCO warns if current levels of greenhouse gas emissions persist, World Heritage coral reef sites may disappear by the end of this century.

27.4.1 Bleaching Events

Corals undergo stress when ocean temperatures change, leading them to expel the algae they host. These algae are responsible for the vibrant colour of the reef, and without them, only the white coral skeletons are visible, resulting in a “bleached” appearance. As corals depend on algae for nutrition, the absence of algae weakens the coral polyp (WWF 2023). In the Gulf of Kachchh and Gulf of Mannar, the coral covering declined by 20–40%, in the Lakshadweep Islands by 20–30%, and in the Andaman and Nicobar Islands by less than 10% as a result of widespread coral bleaching in 1998 (Venkataraman 2011). Venkataraman et al. (2004) reported that only 25% of live corals remained in the Gulf of

Mannar, with branching corals like *Acropora* and *Pocillopora* spp. being the most affected, along with *Montipora*. Marimuthu et al. (2013) and Krishnan et al. (2013) stated that the 2010 bleaching event in the Andaman Islands caused severe bleaching occurrences that were linked to high sea surface temperature. Another catastrophic bout of bleaching struck the coral reefs in GoM and Palk Bay in April and May of 2010, with 85.1% and 87.2%, respectively. The GoK experienced bleaching in 2014–2015, according to Adhavan et al. (2014). This was probably caused by a mix of high sea surface temperature, sedimentation, and a delayed monsoon. High sea surface temperatures of 34.0°C in 2016 prompted bleaching of almost 70% of the corals in Palk Bay and 46% of the corals in the coastal Thoothukkudi Islands in the Gulf of Mannar (Krishnan et al. 2018). The bleaching resulted in a substantial increase in dead coral cover, ranging from 60% to 80%.

Arthur et al. (2005) found that the majority of Lakshadweep's atolls had a consistently low living coral cover. In April–May 2002, Harithsa et al. (2005) measured temperatures in the Kavaratti atoll that were above the bleaching threshold causing coral bleaching. From May to June 2010 an average coral bleaching of 76.5% on Agatti reefs was recorded (Ajith Kumar and Balasubramanian 2012).

27.4.2 Ocean Acidification and Calcification Challenges

Another consequence of climate change is ocean acidification, a process driven by the absorption of excess carbon dioxide by the oceans. Acidification poses a direct threat to the ability of corals to build their exoskeletons through calcification. As the oceans become more acidic, it becomes increasingly challenging for corals to maintain their structural integrity, making them susceptible to physical damage and erosion. The reduced resilience makes them more susceptible to additional stressors, intensifying the overall risk to coral health. In India, the impact of carbon dioxide (CO₂) emissions has led to the warming

and increased acidity of surface waters (Pendleton et al. 2016). A research investigation carried out in the Andaman Sea revealed that ocean acidification has an impact on the growth of coral by diminishing skeletal density. The study emphasized the correlation between the skeletal growth of *Porites* corals concentrating with the surrounding saltwater environment which affects the chemistry of the coral calcifying fluid by connecting skeletal density to it (Mollica et al. 2018).

27.4.3 Extreme Weather Events and Habitat Destruction

Extreme weather occurrences' frequency and severity, such as cyclones and storms, have been amplified by climate change. These events can cause physical damage to coral reefs, breaking apart delicate structures that have been formed over centuries. The Lakshadweep Islands, during cyclones in 1847, 1891, 1922, 1941, and in subsequent occurrences, have witnessed significant coral damage. This has disrupted the fragile equilibrium of the marine ecology. The combined impact of climate change and cyclonic events has had a profound and detrimental effect on these systems, causing impairment of the centuries-old biological structure of coral reefs. As these events increase in frequency, the recovery period for coral reefs is shortened, rendering them more vulnerable to subsequent disturbances (Riyas et al. 2020).

27.5 Restoration and Monitoring

United Nations Environment Programme (UNEP) proposed the term “Coral reef restoration” refers to a series of proactive actions intended to increase the numbers of significant species in addition to strengthening the reef ecosystem's composition and abilities. Coral restoration methods include proceeding coral gardening, moving coral fragments without a transitional nursery phase, larval enhancement, sexually derived coral larvae at a restoration site, adding artificial structures such as shipwrecks for

the purpose of restoring coral reefs and stabilizing the substratum to aid in coral recruitment or recovery. The report says that the restoration methods are not equivalent to more traditional approaches such as threat reduction and marine protection. On the other hand, long-term success with restoration might depend on its combination with vigorous local conservation initiatives and measures to alleviate the impact of climate change.

Coral restoration sceptics contend that it is unproductive unless it can rebuild reefs at the ecosystem level and diverts priority from resolving climate change and other problems to the marine environment. In response, proponents of coral restoration argue that although large-scale problems like coastal pollution and climate change are being mitigated, short-term interventions can help preserve coral biodiversity.

27.5.1 India in Reef Restoration

The Ministry of Environment Forest and Climate Change (Government of India), the Forest Department (Government of Tamil Nadu) partnered with Suganthi Devadason Marine Research Institute (SDMRI) in 2002 to start coral regeneration programme along the Tuticorin coast of the Gulf of Mannar (CORDIO). The affordable, low-tech and low-cost transplantation technique that the SDMRI researchers used to continue coral restoration using artificial substrates was successfully standardized and field tested in the region. The Tuticorin coast has restored more than 1 km² of deteriorated reef area, with an overall survival rate of over 80%.

Due to increased temperatures and other reasons, the GoM coral reef colonies experienced significant bleaching during 2005, 2010, and 2016. So the National Centre for Coastal Research (NCCR), the Ministry of Earth Sciences, Government of India started a coral restoration project which resulted in the recovery of live coral cover in dead coral reef areas in the Gulf of Mannar islands. NCCR was also successful in restoring Lakshadweep region corals, viz.

Agatti and Kavaratti in an area, approximately of one acre size. The activity covered transplantation with branching corals, massive corals, and foliaceous corals. Department of Science and Technology (DST), Lakshadweep Islands, helped in this activity. DST also actively supports the research related to the discovery that shipwrecks at the Vembar (Gulf of Mannar), Zuis (Goa), North Bay (Port Blair, South Andaman), Peel (Havelock, South Andaman), and Japan (Car Nicobar) sites produced new, productive ecosystems with higher biodiversity (Yogeshkumar et al. 2015).

The Central Marine Fisheries Research Institute (CMFRI, Government of India) provided technical support to the Government of India for the installation of artificial reefs offshore in Chennai and nearby regions. In a record of 45 days, 6000 artificial reef units totalling about 5250 metric tonnes were transported in fishing vessels and put into place; this effort has begun ushering marine life. Similar work of deployment of the artificial reef has been undertaken by Temple Adventure in Puducherry waters and found positive outcomes.

With assistance from Tata Chemicals Limited, the Gujarat Forest Department and Wildlife Trust of India collaborated to launch the 2008 Coral Reef Recovery Project. A portion of a marine region is being actively maintained by a community for the first time in India. Situated 12 km south of the Gulf of Kachchh in Gujarat, it aims to create and carry out suitable plans for the preservation of the Mithapur Reef (Fig. 27.3).

Further, a dead coral reef was found by Indian marine scientists from the Zoological Survey of India (ZSI) which ages to thousands of years old in the Gulf of Kachchh and decided to take an attempt to revive it. In addition, ZSI is implementing a first-time technique for revitalizing coral reefs utilizing mineral accretion technology, or biorock, with assistance from Gujarat's Forest department. In 2019, the installation of a biorock structure was done in the Gulf of Kachchh, one nautical mile off the shore of Mithapur. Because of the large tidal amplitude in GoK, 4-m biorock was installed at the site where



Fig. 27.3 Coral reef restoration in the GoK, west coast of India. (Photo Credit @ Venkatraman, K., ZSI “Restoring Coral Reefs at GoK”. 2015)

the region has low tide depth at high tide. A total of 1000 m² coral transplantation was successfully completed by ZSI in three different places in the GoK, i.e., Mithapur reef, Narara reef, Pirotan Island, at Gulf of Kachchh from 2013 to 2015 and found *Acroporidae* (*Acropora* sp. and *Montipora* sp.) to be suitable for large-scale transplantations (Yogeshkumar et al. 2016). The programme was funded by the Society of Integrated Coastal Management, Ministry of Environment Forest and Climate Change (MoEFCC) through the World Bank-Integrated Coastal Zone Management Project (ICZMP), Gujarat Ecology Commission (GEC) in this region.

Additionally, Reef Watch launched the RE (EF) Generate programme in 2018 to restore coral in the Andaman Islands through the use of Mineral Accretion Technology. A restored reef patch can be adopted through the Reef Guardian

Programme with the objective to promote a personal link between mankind and the reef.

The Intergovernmental Panel on Climate Change (IPCC) warns that coral reefs are at serious risk as global temperatures rise. If the planet warms by 1.5 °C, an estimated 70–90% of coral reefs could disappear. At 2 °C of warming, over 99% are expected to be lost. The global temperature is currently 1 °C higher than it was before industrialization. If we do not take coordinated action to reduce emissions and encourage adaptation and faster recovery from injury, we face the very real prospect that the world's coral reefs as we know them could disappear within a generation. This issue will be made worse by the growing frequency of disturbances that affect coral reefs since there is insufficient recuperation time between the distressing occurrences. In this context, protecting coral reefs necessitates effective reef restoration efforts.

27.5.2 Monitoring Techniques

Monitoring coral reefs is crucial for understanding their health and resilience. Growing recognition of the extensive threats confronting coral reef ecosystems has underscored the importance of monitoring activities in evaluating the repercussions of disturbances on reefs and monitoring subsequent recovery or degradation. Numerous coral reef monitoring initiatives relying on field surveys have been established to gauge the global coral reefs' health condition.

27.5.2.1 Implications of Geospatial Technology to Study the Coral Reefs of India

Satellite earth observations and geographic information system (GIS) based analytical capabilities have revolutionized the mapping and monitoring of coral reefs. Traditionally, coral reef classification has been attempted, involving grouping coral reefs into different classes based on field observations. In the last few decades, technological advancements in geospatial techniques improved the mapping and monitoring of coral reefs (Kennedy et al. 2021). Remote sensing offers synoptic coverage for inferring large-scale patterns in coral reefs and complements field surveys for studying the reefs' spatial, spectral, and temporal dynamics (Hedley et al. 2016). Initially, reefs were merely interpreted using satellite images in the 1980s, just like identifying the reefs from field photographs (Fig. 27.4), and it has been extended to mapping habitats that are conducive to reef development (Fig. 27.5). In the early 2000s, the accessibility to higher spatial-spectral resolution images and computer-based processing techniques enabled a detailed mapping of reefs with physical attributes. Optical sensors that function within the visible range of the electromagnetic spectrum (400–700 nm) can penetrate up to 25 m deep in clear waters and provide valuable information about shallow reef platforms (Hamylton 2017). In addition, the near-infrared (NIR) band of various satellite sensors is potentially helpful for coral reef researchers (Andréfouët et al. 2003). Indexes like the

Normalized Differential Vegetation Index (NDVI) are constructed using various combinations of visible and NIR bands, which provide valuable signatures for identifying environmental conditions that affect reef development. To strengthen coral reef resilience in the face of global warming, NOAA launched the Coral Reef Watch programme in 2000, utilizing near real-time satellite-derived products on a global scale (Fig. 27.6). However, the satellite-based approaches are limited by the data availability and resolution trade-offs. Obtaining remote sensing data for a coastal region with high spatial and spectral resolutions is difficult under cloud-free conditions and repeated coverage (Kennedy et al. 2021).

The remote sensing data with comprehensive information on reef systems is the first World Atlas of Coral Reefs published by UNEP World Conservation Monitoring Centre (Spalding et al. 2001; Hamylton 2017). In recent years, geospatial scientists have overcome these shortcomings by developing automated methods for mapping the coral reefs in conjunction with spectral reflectance and bathymetry information. The emerging technologies, including 3D mapping, autonomous underwater vehicles (AUVs), drones, and machine learning tools, have revolutionized mapping and modelling capabilities and helped coral reef conservation communities (Madin et al. 2019). For instance, scientists, universities, NGOs working in coral reefs, and private entities published the Allen Coral Atlas which consists of mapping of the world's coral reefs and their threats in unprecedented detail using high-resolution satellite imagery and advanced analytics (Fig. 27.7).

Similar to these world scientific trends, in India, the coral reefs were studied scientifically, from traditional field-based classification to modern, robust methods, by a diverse group of scientists. All major coral reef areas in India, including the ANI, LoI, GoK, and GoM, were studied using a convergence of geospatial techniques which revealed the current status of coral reefs, changes in their spatial extent and coverage, and life-threatening aspects/events (Fig. 27.4). The stud-

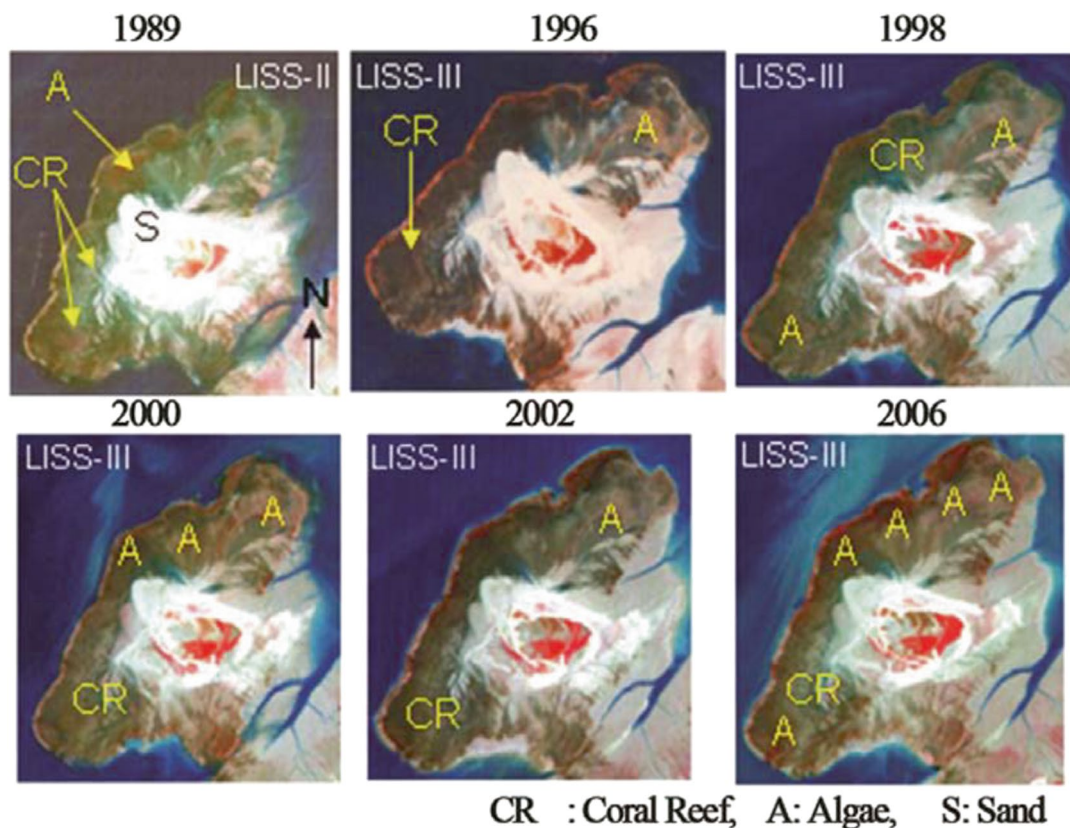


Fig. 27.4 Images from the IRS satellite series LISS-II and LISS-III sensors reveal the successive degradation of Pirotan Reef, Gulf of Kachchh, which

covers both coral and macroalgae-dominated regions during 1989–2006. (Source: <https://nopr.niscpr.res.in/handle/123456789/24796>)

ies are also extended to document various natural and anthropogenic stresses that harm the development and sustenance of coral reefs in India.

The recent availability of multi-spectral datasets and developments in digital image classification methods of coral reef monitoring significantly improved in India. The ongoing free provision of Landsat and IRS-Resourcesat, satellite datasets for Indian regions, helps scholars and agencies study diverse threats the coral habitats face (Bahuguna et al. 2013). Nimalan et al. (2021) study in the GoM region found that spectral peaks around 485 nm and 590 nm wavelengths easily distinguished live coral species from dead or bleached corals; they found that bleached and dead coral reefs do not have peaks in these spectral wavelengths. Many studies attempted in India link remote sensing-derived environmental

factors with reef distribution, allowing the scientists to characterize site-specific features of the Indian coral reefs and perform physical and biological linkages of them at local, regional, and global scales.

Bahuguna et al. (2013) inventorised the coral reef habitats of the Central Indian Ocean using multi-temporal satellite datasets (2004–2007) and found that GoK and GoM reefs are in degraded condition due to coral bleaching due to anomalous rise in SST, the effect of sedimentation and anthropogenic pressures. This revealed the coral reefs in the LoI enjoy pristine ecological status to some extent, while the coral reefs in the ANI were in pristine ecological conditions before the 2004 Indian Ocean Tsunami. The deadly waves damage about 54% of Andaman reefs. The modern assessments show that LoI and ANI reefs

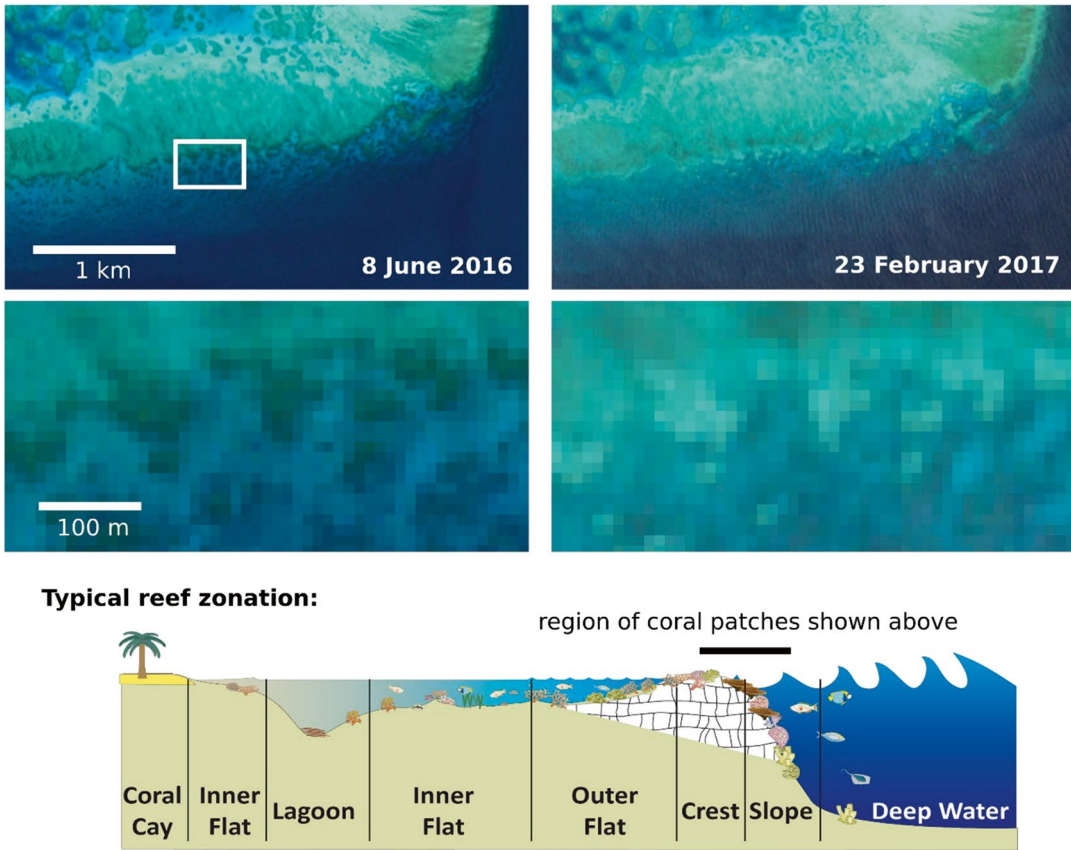


Fig. 27.5 Image of coral turning bright white in Adelaide Reef, Central Great Barrier Reef generated from the Copernicus Sentinel-2A satellite on 8 June 2016 and 23

February 2017. (Source: https://www.esa.int/ESA_Multimedia/Images/2017/05/Sentinel-2_captures_coral_bleaching)

are vulnerable. The satellite images also reported the coral bleaching events between 1998 and 2016, with severe impacts in 2010 in Andaman Island reefs. Also, Majumdar et al. (2018) recently used SENTINEL-2A data to report the distribution of coral reefs in the Andaman Islands. With the help of knowledge-based classification algorithms, they found that the Andaman Islands' coral reefs are affected by natural hazard events, including earthquakes, land subsidence, and tsunamis. Their study also reported the coupling mechanism of elevated SSTs and El Niño that cause coral damage in the Andaman Islands.

In addition to satellite images, GIS provides a promising tool for analysing environmental factors to characterize coral reef habitats along the Indian coasts. Databases such as MARSPEC and Bio-ORACLE (Sbrocco and Barber 2013; Assis

et al. 2017) provide pelagic and benthic marine layers to model the present and future conditions of the reefs using GIS. Using these datasets, Ellepola et al. (2021) modelled in a GIS environment and found that a higher coral depletion is expected in the GoM. GIS-based modelling studies help to understand the ecological settings for conserving unique reef habitats around the country. Implementing geospatial technologies with optical, microwave, hyper-spectral, and acoustic sensor datasets through GIS is essential for monitoring and managing coral reefs in India.

27.5.2.2 Environmental DNA (eDNA) and Its Role in Coral Biodiversity Monitoring

The utilization of eDNA gained interest in various biological disciplines, encompassing terres-

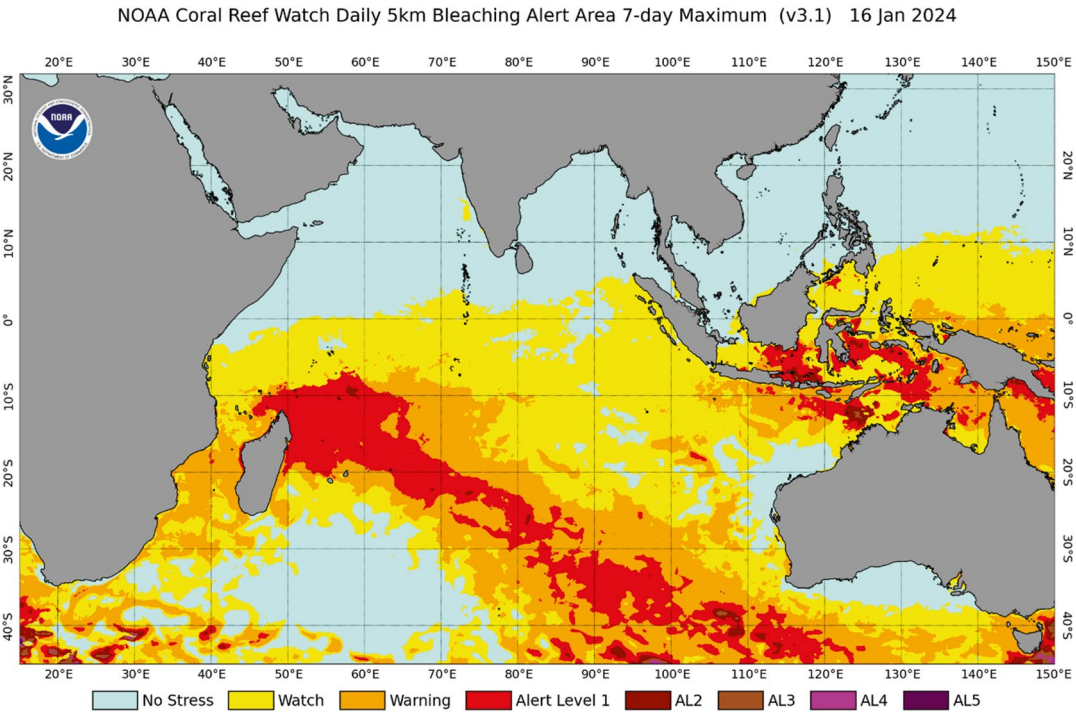


Fig. 27.6 NOAA coral reef bleaching alert product based on CRW's satellite SST monitoring during 10–16 January 2024 and updated daily at a 5 km satellite data grid. (Source: <https://coralreefwatch.noaa.gov/product/5km/index.php>)

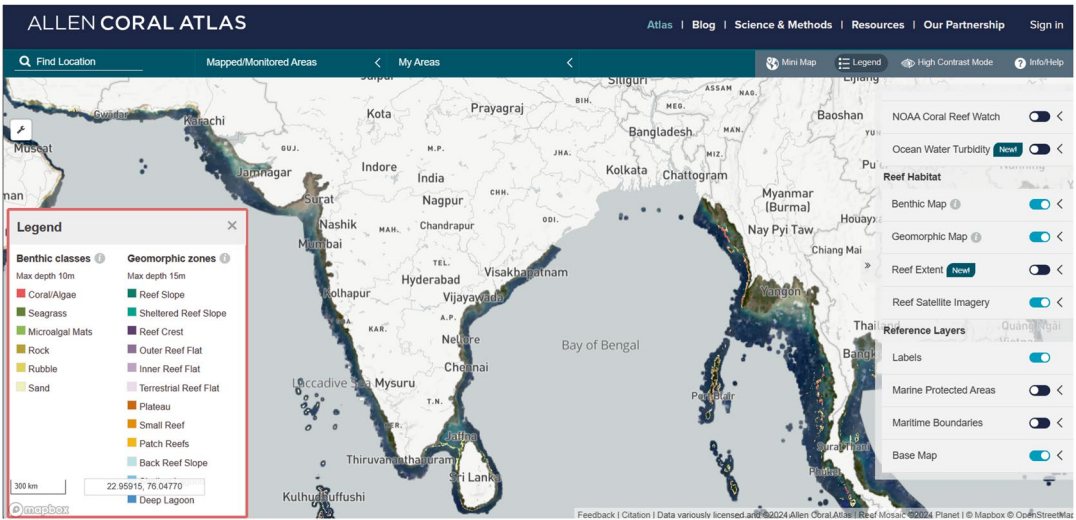


Fig. 27.7 The distribution of coral reefs and their geomorphic zones, Indian sub-continent by the Allen Coral Atlas, which maps and monitors coral threats using a

high-quality (3.125 m) mosaic from Planet's Planet Scope satellite imagery. (Image credit: Allen Coral Atlas <https://allencoralatlas.org/>)

trial and aquatic ecology, conservation, public health, and ecosystem engineering. eDNA is based on the principle that the DNA of each individual organism can be retrieved from any given environment. This DNA is released through excretion, shedding of body parts, mucus, egg and sperm deposition, and decomposed tissue scraps after death (Ogden 2022). While the monitoring of fish species through eDNA is well-established, especially in invasive species monitoring, recent research, like that of Gelis et al. (2021), has successfully employed eDNA metabarcoding to accurately identify coral reef-associated fish species and reveal higher species diversity in specific reef zones. However, assessing coral distribution and health, which is vital in the ongoing climate change, remains a complex task. Nichols and Marko (2019) found a robust correlation between visual estimates of coral

cover and eDNA abundance in Hawaii reefs, highlighting the potential of eDNA to monitor coral biomass. Additionally, the eDNA method detected more coral genera compared to visual surveys, with long reads proving more effective in detecting a broader range of coral genera (Nichols and Marko 2019). Coral species along with other 14 eukaryotic community members were detected using eDNA metabarcoding with 18S rRNA eukaryotic primer from the lagoon and reef slopes of Ningaloo coast of Australia (Fig. 27.8) (Dugal et al. 2023).

Alexander et al. (2020) developed a comprehensive eDNA toolkit designed for monitoring coral diversity and validated its performance against visual surveys. Everett and Park (2018) showed that eDNA sequencing can complement traditional surveys by identifying deep-sea octocorals that are challenging to visually identify.

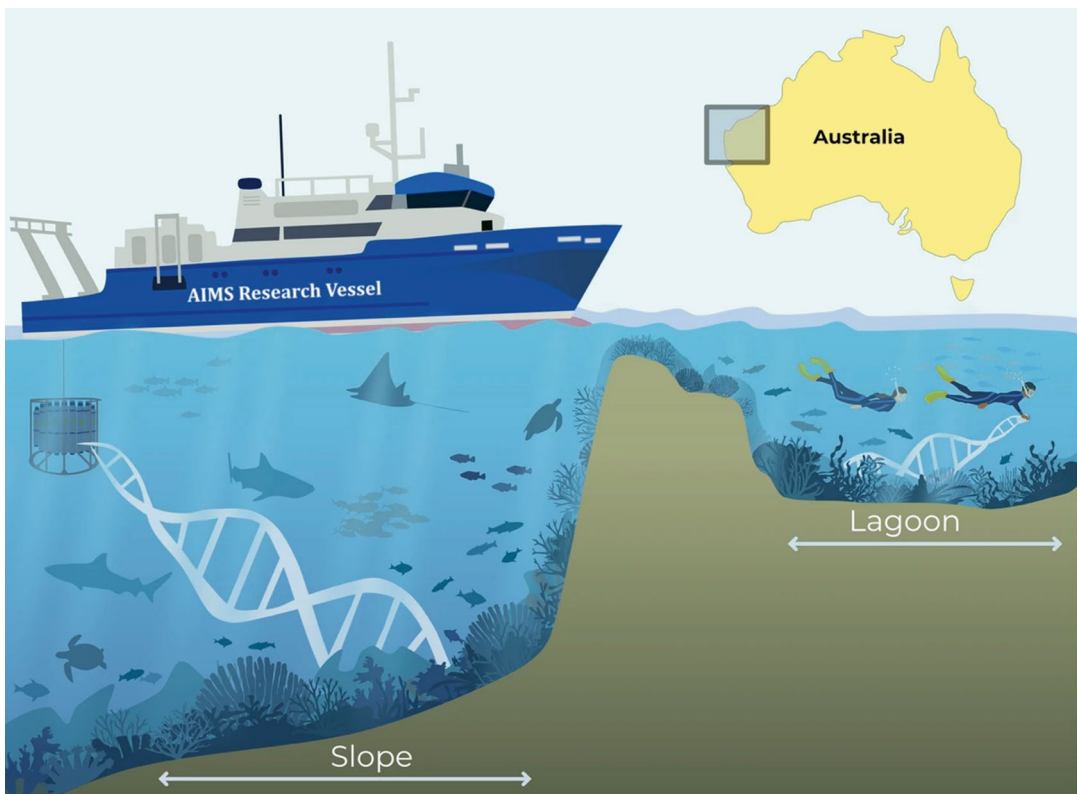


Fig 27.8 Dugal et al. (2023) use of eDNA to identify the coral community from the environments of the reef slopes and lagoons of Western Australia's Ningaloo Coast World Heritage Area. (© Dugal, L., Thomas, L.,

Meenakshisundaram, A. et al. Distinct coral reef habitat communities characterized by environmental DNA metabarcoding. *Coral Reefs* 42, 17–30 (2023). <https://doi.org/10.1007/s00338-022-02301-3>

Overall, these findings highlight the potential of eDNA as a non-invasive and efficient method for coral assessment. Kutti et al. (2020) used eDNA, and primer targeting 178 bp of *Lophelia pertusa* (cold-water coral (CWC) reefs) mitochondrial DNA, and successfully detected elevated eDNA concentrations near large reefs in Norwegian fjords, particularly those on surface water. Kutti et al. (2020) successfully modelled eDNA dispersal in the well-studied Norwegian fjords, including CWC reefs in the region. The model demonstrated high accuracy, in measuring eDNA concentrations in the water (Fig. 27.9). This cost-effective method offers a rapid way to locate challenging-to-detect “wall reefs,” presenting a promising solution to supplement traditional mapping techniques (multibeam echosounders and ROVs), and costly surveys in identifying vulnerable marine ecosystems.

Beyond the well-recognized threats of climate change and rising water temperatures, another crucial facet of coral reef conservation involves addressing concerns related to coral predators. Sea stars, fish, marine worms, barnacles, crabs, and snails are among the predators that, during outbreaks, can inflict significant damage on coral

reefs. In a study by Uthicke et al. (2018), environmental DNA (eDNA) and a PCR-based method were employed to identify crown-of-thorns sea stars in the Australian Great Barrier Reef. The droplet digital (dd) PCR method efficiently identified sea star DNA both in laboratory and field settings, with the identified sea star population through ddPCR showing correlation with visual surveys. This approach utilizing eDNA and ddPCR proves valuable in monitoring predator populations in reef areas.

The application of environmental DNA (eDNA) and metabarcoding, with targeted mitochondrial COI gene primers specific to marine biodiversity as demonstrated by Levy et al. (2023), proved effective in detecting a wide range of marine organisms. This initiative established a foundational biodiversity baseline, offering data on diversity, richness, and species abundance in the Gulf of Eilat/Aqaba. The study highlighted the capacity of eDNA to reveal cryptic organisms.

Beyond recording species richness and diversity at a high taxonomic level, eDNA was employed to identify corals at the genus level. In the work by Nishitsuji et al. (2023), over 43

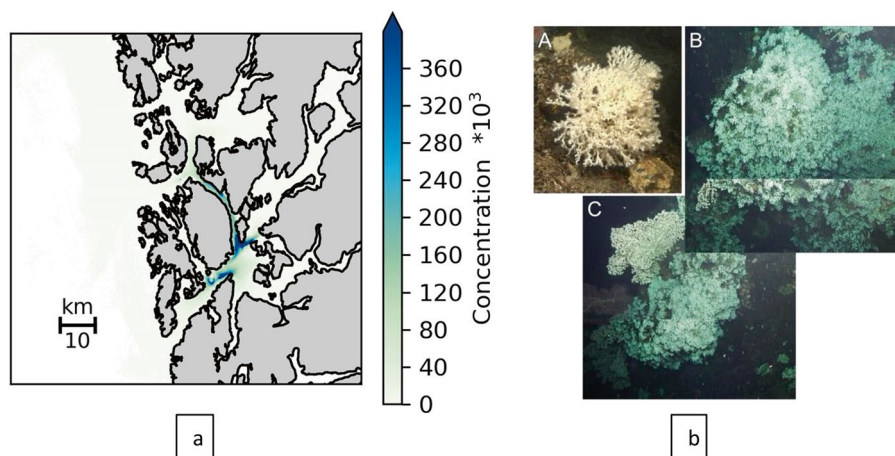


Fig. 27.9 (a) eDNA-based dispersal study was conducted in May 2018 from 200 m depth (Nakken CWC reef) using the ROV Aglantha. The simulation shows the dispersal of *Lophelia pertusa* genetic material was decreased with increase in distance from the coral reef. After 12–31 km from the reef only 5% of the coral genetic material is detectable. (b) *Lophelia pertusa* colonies in

different sizes: (A) 25 cm in diameter, (B) 280 cm high and 350 cm wide, and (C) 300 cm high and 180 cm wide. (Kutti T, Johnsen IA, Skaar KS, Ray JL, Husa V and Dahlgren TG (2020) Quantification of eDNA to Map the Distribution of Cold-Water Coral Reefs. Front. Mar. Sci. 7:446. <https://doi.org/10.3389/fmars.2020.00446>)

Scleractinia genera were identified from surface seawater above reefs near Okinawa Island using the eDNA approach. The correlation between eDNA results and direct conformation by divers confirmed the presence of major genera, including *Acropora*, *Montipora*, *Pocillopora*, and *Porites* (Fig. 27.10). Notably, eDNA detected more coral genera than visual survey and revealed the undocumented species. This specific eDNA

method for *Scleractinia* corals holds promise as a robust and powerful tool for comprehensive surveys of coral reefs.

Similarly, with the use of in-house fusion primer, Gösser et al. (2022) identified 29 Scleractinian coral genera belonging to 14 families from the seawater of Koh Pha-ngan, Thailand. Similar to the findings from other studies, the eDNA survey indicates a log-linear correlation

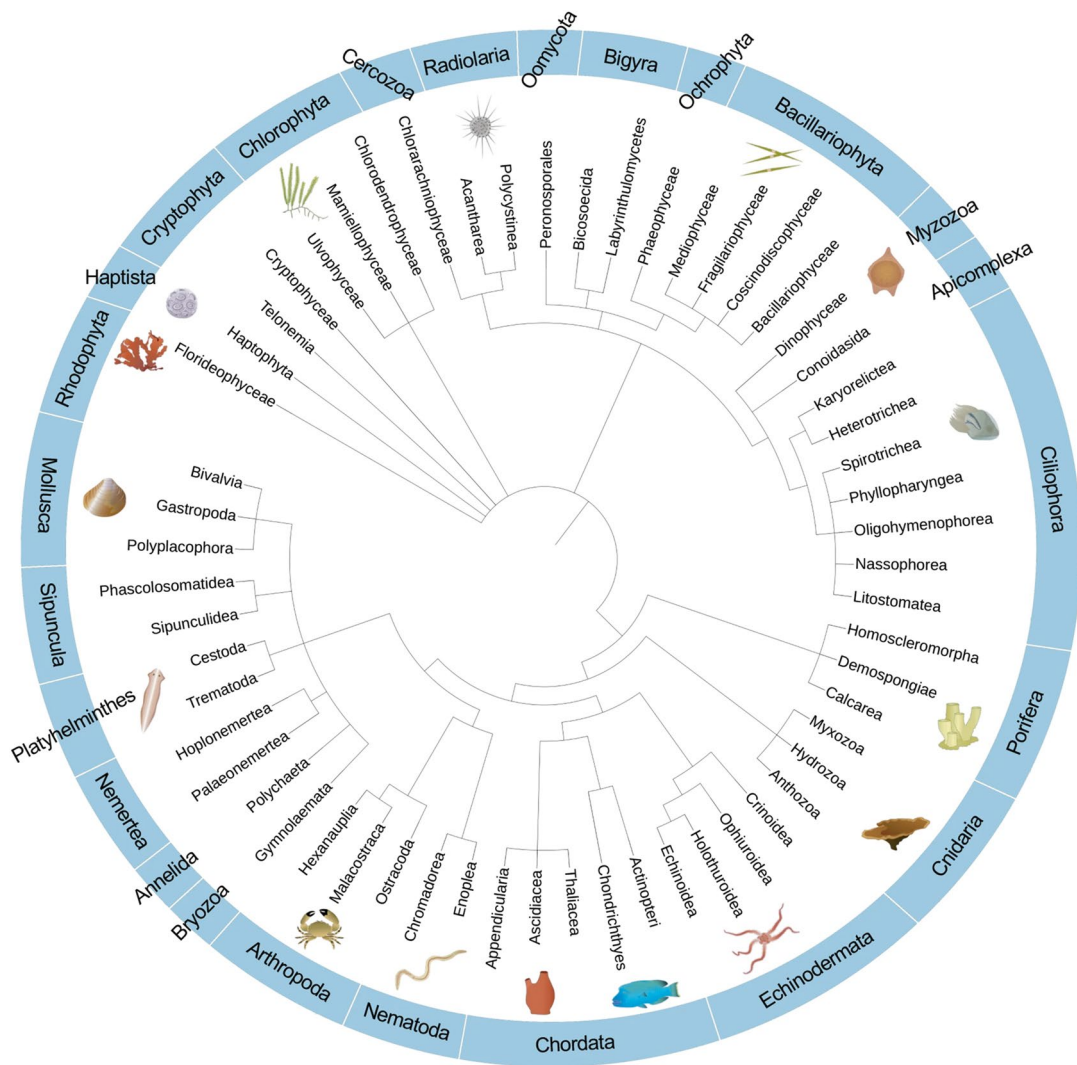


Fig. 27.10 Based on the eDNA, rRNA 18S Universal eukaryote and amplicon sequencing Dugal et al. (2022) identified the major eukaryote organisms like Protozoa, Cnidaria, Chromista, Apicomplexa, Radiolaria, Ciliophora, etc. (Dugal, L., Thomas, L.,

Meenakshisundaram, A. et al. Distinct coral reef habitat communities characterized by environmental DNA metabarcoding. *Coral Reefs* **42**, 17–30 (2023). <https://doi.org/10.1007/s00338-022-02301-3>)

between eDNA read abundance and visual estimation at the genus level suggesting a predictive correlation between eDNA reads and coral coverage. Besides, a variation in eDNA reads between diurnal and nocturnal samples was noted, which correlates with the activity of coral phases. Notably, the use of labelled fusion primers yielded similar results compared with commercial kits. This indicates that the utilization of different primers might enhance the large-scale study and help to increase the library which researchers may explore further. These findings are crucial to establish eDNA as a supplemental method alongside visual surveys for monitoring coral reefs.

27.6 Conservation Measures

The international community has committed to implementing coordinated policy measures to protect coral reef ecosystems. Various United Nations (UN) conventions play a significant role in restoration efforts, including the United Nations Convention on the Law of the Sea (UNCLOS) and the United Nations Framework Convention on Climate Change (UNFCCC), along with the Paris Agreement. Moreover, the United Nations Environment Programme (UNEP) supports coral restoration through four specific conventions: (1) the Conservation of Migratory Species of Wild Animals (CMS), (2) the Convention on Biological Diversity (CBD), (3) the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), and (4) the Regional Seas Conventions and Action Plans. Additionally, the United Nations Educational, Scientific and Cultural Organization (UNESCO) oversees two conventions dedicated to these efforts: the Ramsar Convention on Wetlands and the World Heritage Convention (UNWHC).

The UN Resolution on the Decade on Ecosystem Restoration aims to “support and scale up efforts to prevent, halt, and reverse the degradation of ecosystems worldwide and raise awareness of the importance of successful ecosystem restoration.” UNEP now extends support

to compile a volume on best practices for coral reef restoration. Additionally, the UN Decade of Ocean Science for Sustainable Development (2021–2030) will enhance the Decade on Ecosystem Restoration, as scientific understanding is crucial for successful reef restoration.

Two significant global partnerships include the International Coral Reef Initiative (ICRI) and the International Union for Conservation of Nature (IUCN) through The Restoration Initiative (TRI), which work to identify and address the pressures on coral reefs. Furthermore, various prominent organizations such as the Coral Restoration Foundation and the NOAA Coral Reef Conservation Program are actively engaged in reef restoration activities worldwide. Approximately, 35 million people, in India, live within a distance of 30 km from the coral reef, and most of the coastal communities depend on reefs for their livelihoods and as a source of edible protein. Various legal frameworks have been enforced, including the Wild Life (Protection) Act (WLPA) 1972, Marine Fishing Regulation Act (MFRA) 1983, 2000, and the Coastal Regulation Zone (CRZ) Notification, 2011, to protect and conservation of Indian coral reefs. India has 31 Marine Protected Areas (MPAs), five of which have been surveyed and identified for coral reef protection. Recommendations have been made to halt all human activities that cause mechanical damage, coral removal, and the harvesting of marine organisms, except for scientific research (De et al. 2017). This includes the Marine National Park and Marine Sanctuary in the Gulf of Kachchh; the Malvan Marine Sanctuary in Maharashtra; the Gulf of Mannar Marine National Park in Tamil Nadu (part of the Gulf of Mannar Biosphere Reserve); various national parks, sanctuaries, and biosphere reserves in the Andaman and Nicobar Islands; and the Pitti Bird Sanctuary in the Lakshadweep Islands. These areas provide legal protection for coral reefs and their biodiversity, with some protections in place since the 1970s. Overall, coral reef ecosystems in India are granted the highest legal protection status to ensure their long-term preservation and conservation.

27.7 Conclusion

The detailed examination of India's coral reefs unveils their rich biodiversity and essential role in supporting coastal ecosystems and livelihoods. Despite their significance, these invaluable ecosystems confront a myriad of threats, encompassing the impacts of climate change, local stressors, and a consequential decline in reef health. Coral bleaching, ocean acidification, extreme weather events, and habitat destruction have resulted in substantial losses in live coral cover across diverse regions.

Addressing these challenges and conserving India's coral reefs involves a comprehensive strategy. International commitments, legal protection frameworks, conservation initiatives, artificial reef deployments, community-managed conservation, and government support all play crucial roles in safeguarding these fragile ecosystems. The integration of technologies such as geospatial tools and environmental DNA (eDNA) presents promising avenues for monitoring, assessing, and understanding coral reef status.

These collective efforts underscore the urgency of tackling both global and local stressors to ensure the survival and resilience of India's coral reefs. Conservation measures, community involvement, scientific research, and policy interventions are pivotal for the sustained protection and restoration of these diverse and economically significant ecosystems. In conclusion, coordinated endeavours, including international collaborations, the use of modern technological tools, and community engagement, are imperative to preserve India's rich coral biodiversity, mitigate threats, secure the future of these invaluable ecosystems for future generations, and contribute to sustainable marine resource management.

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